

mdstats 0.20.181a0 Patch Notes

PERF-P2 lazy TARGET-DATA2C authority v2

mdstats project

2026-08-15

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This release closes MLFF gate **PERF-P2**.

Changed

- Advances TARGET-DATA2C plan authority from v1 to v2.
- Materializes and scores nested target-size rungs lazily.
- Stops only after the active TARGET-DATA2D shortlist width is globally fixed.
- Records explicit global rung qualifications, intentional non-materialization, pool-unavailable sizes, stop reason, and monotonicity contract.
- Keeps legacy v1 readable as a regression oracle while rejecting it as stale current campaign authority.
- Makes exact single-rung cKDTree worker count execution-only and digest invariant.
- Updates TARGET-DATA2D to consume v2 materialized qualification evidence without treating intentionally absent larger rungs as failures.

Qualification

Exhaustive v1 and lazy v2 produce identical Stage-A survivor sizes and exact survivor evidence on both exhaustive-fallback and forced-early-stop fixtures. On the supplied-data-derived early-stop fixture, three fresh-process samples show an 80.23% median wall-time reduction and an 87.50% reduction in serialized ladder authority size.

The fallback timing is intentionally not claimed as a speedup because its observed ranges overlap and are cloud-scheduler sensitive.

Next

PERF-P3 - CPU structural and reduction hardening.